REINDEXREPAIR: STATE-CONSISTENT EVALUATION OF POISON REPAIR IN RETRIEVAL-AUGMENTED GENERATION

ABSTRACT

Deleting evidence in retrieval-augmented generation (RAG) need not repair the rebuilt service: source edits may expose backup poison or migrate residual text across chunks. We formalize this state mismatch as *repair mirage* and introduce REINDEXREPAIR, a materialized re-index protocol with exact and provenance-complete controls. On frozen Natural Questions, target recurrence is 13/71 after trace-visible deletion versus 2/71 after complete-lineage removal (paired difference 15.5 points; Holm-adjusted $p = .0020$). A separate boundary mechanism yields 7/18 versus 1/18 ($p_{\mathrm{Holm}} = .0312$) in a boundary stress setting. In an exhaustive 30-incident oracle, REINDEXREPAIR finds a safety-and-correctness-preserving edit for 19/30 cases; after four-way validation, exact-view finalist recall is 19/19. This is an offline human-oracle analysis, not a deployable selector. Under adjudicated labels, the results support lineage-complete controls in these constructed settings; least-favorable endpoint assignments retain positive differences but not statistical significance.

***Index Terms*—** retrieval-augmented generation, data poisoning, counterfactual evaluation, re-indexing, information forensics

1. INTRODUCTION

Retrieval-augmented generation (RAG) couples language generation with external evidence [1, 2]. PoisonedRAG shows that a few injected passages can steer answers in million-scale corpora [3]. Removing evidence behind one failure, however, need not repair the next execution.

A repaired source is normalized, tokenized, chunked, embedded, indexed, retrieved, reranked, and prompted again. Removing a top-ranked poison can promote a hidden copy, while re-chunking can migrate residual text into a new retrievable unit. Dynamic-index work similarly distinguishes logical deletion, physical deletion, and rebuilding [4]. We call false assurance caused by this state mismatch a *repair mirage*.

The central innovation is to make the *reconstructed serving state*, rather than deleted evidence, the unit of repair validation. Our contributions are threefold. **(i) State-consistent formulation:** we define answer-level repair mirage and a root-lineage closure that invalidates every old descendant of an edited source. **(ii) Mechanistic paired evidence:** fresh-index interventions separate hidden-copy promotion and boundary migration from stale vectors, with blinded human adjudication and matched provenance-complete controls. **(iii) Checkable counterfactual evaluation:** an exact view reuses only validated unchanged state and is certified against separately materialized rebuilding; an exhaustive human oracle quantifies feasible minimum-token edits and abstentions; matched utility tests quantify retrieval effects. The protocol is not a deployable locator or semantic selector and makes no universal safety claim.

Table 1. Contrasting evaluation states and their primary endpoints.

Line of work	State evaluated	Primary endpoint
Traceback / filtering	Attribution / served context	Localization / attack rate
ANN deletion	Mutated vector index	Recall / latency / memory
REINDEXREPAIR	**Rebuilt serving path**	**Paired answer recurrence**

Data and code availability. Reproduction code, de-identified evaluation artifacts, and source-data reconstruction instructions are available at `github.com/violetzxd/ReIndexRepair`. Public corpora and model weights remain under their original licenses.

2. RELATED WORK

Document attribution [5, 6] and span traceback [7] localize evidence in an observed execution; retrieval defenses filter chunks [8, 9] or graph knowledge [10]. Pipeline-aware attacks [11] and explicitly chunk/reranker-aware poisoning [12] study poison effectiveness in the serving pipeline. Behavioral unlearning and knowledge erasure instead seek forgetting [13, 14]. These objectives differ from our matched comparison of trace-derived and provenance-complete source repairs after reconstruction. ANN-deletion evaluation measures index quality and systems cost [4], rather than answer recurrence (Table 1). We complement these lines with a source-to-answer intervention protocol, not a claim of superior localization, filtering, or forgetting.

3. REINDEXREPAIR METHODOLOGY

Figure 1 separates trace diagnosis, exact counterfactual evaluation, and source-materialized validation. Every branch uses a frozen pipeline; human feasibility is assessed only after all candidate outputs exist. **State model and repair objective.** Let D be versioned source documents and let the frozen pipeline fingerprint θ identify every normalizer, tokenizer, chunking rule, embedding and reranking model, prompt, and decoding parameter. Its materialized retrieval state is

$$\mathcal{I}_\theta(D) = \mathrm{Index}_\theta(\mathrm{Embed}_\theta(\mathrm{Chunk}_\theta(\mathrm{Norm}_\theta(D)))). \quad (1)$$

The output $Y(q; D, \theta)$ also includes retrieval, reranking, prompt assembly, and generation. A source action a produces D_a. A repair claim is state-consistent only if it is evaluated through the same

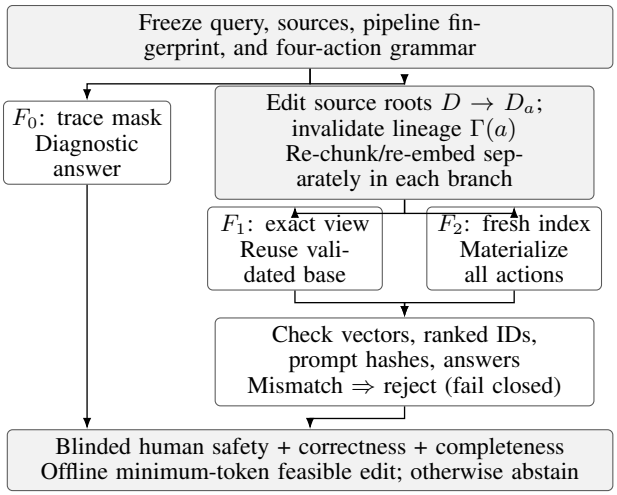


Fig. 1. Offline repair-lattice workflow. F_0 diagnoses one trace; F_1 and F_2 evaluate edited source states. All four actions are run in F_2 before human-oracle selection; certificate failure rejects a candidate.

$D_a \mapsto \mathcal{I}_\theta(D_a) \mapsto Y(q; D_a, \theta)$ path that will serve after deployment. Prompt masks and vector tombstones can diagnose one trace, but by themselves cannot establish this claim unless their counterfactual state is shown equivalent to a source-materialized rebuild.

Two distinct evaluation questions. The mechanism study pairs trace-derived repair with incident-provenance-complete quarantine, both in F_2, to test recurrence after an apparently safe trace repair. The repair lattice instead compares F_1 with F_2 for each of four source edits, then selects a minimum-token human-feasible action. Provenance closure is a matched control in the first study; exact-view equivalence is a certificate in the second. Neither implies perfect localization.

Root-level lineage manifest. For each source revision r, the frozen pipeline records a materialization map $L_\theta(r)$ containing its descendant chunk and vector-entry IDs. For the source roots $A(a)$ touched by action a, the invalidation set is

$$\Gamma(a) = \bigcup_{r \in A(a)} L_\theta(r). \tag{2}$$

Closure is essential because a deletion can change token offsets and chunk boundaries beyond the visible sentence. Operationally, a touched source root is the seed: we re-normalize and re-chunk the entire affected document, invalidate its recorded old descendant IDs, and attach new descendants to the repaired revision. This conservative root seed avoids mixing span and chunk node types. The trace-derived arm chooses roots only from incident evidence visible in the triggering top-5; the matched oracle control chooses every incident-owned root. Both arms then execute the identical rebuild code path; they differ only in provenance coverage.

For example, deleting a sentence near a 128-token boundary can shift every following chunk even if most characters are unchanged. Invalidating only the visible chunk would then mix an old partition with a new source revision. Root-level invalidation avoids this hybrid state: all old descendants of a touched document are removed before any new descendants enter the view. Conversely, an untouched backup document retains its validated vectors and can rise in rank when competing poison is removed. The protocol therefore accounts for both changed text and changed competition among unchanged text; re-embedding just the edited span cannot represent both effects.

Three views of one action. F_0 edits the logged prompt and asks whether the action *appears* safe on the observed trace. F_1 constructs an exact counterfactual retrieval view without committing a new physical index. F_2 materializes the repaired corpus from source through generation and is authoritative. For target-adoption indicator h_t, a repair mirage is

$$M(a, q) = \mathbb{1}[h_t(Y_{\text{trace}}(q; a)) = 0 \ \wedge \ h_t(Y(q; D_a, \theta)) = 1]. \tag{3}$$

In the repair lattice, F_0 is explicitly a trace-local surrogate: it masks the target string for span deletion, the first served incident chunk for either sentence-level action (the hidden backup is unavailable to the trace), and all served incident chunks for quarantine. It is not an action replay. We report both unconditional F_2 target adoption over frozen incidents and the operational false-assurance rate conditional on initial target adoption and an F_0-safe response. This conditioning is fixed before either rebuilt endpoint is read, and the incident remains the statistical unit.

Counterfactual construction. Let E_0 denote the labeled vectors in the frozen index and D_a^{aff} the affected repaired documents. F_1 forms

$$\widetilde{E}_a = (E_0 \setminus \Gamma_E(a)) \cup \text{Embed}_\theta(\text{Chunk}_\theta(\text{Norm}_\theta(D_a^{\text{aff}}))), \tag{4}$$

where $\Gamma_E(a)$ is the vector projection of the closure. It scores the frozen query vector against every member of $\widetilde{E}_a$, orders equal scores by chunk ID, applies the frozen reranker and prompt builder, and decodes greedily. Unchanged descendants are reused only after validating their ID, text, and vector fingerprint bundle; every changed source is separately re-chunked and re-embedded in each materialization. Thus the view exposes promoted backup poison and boundary-created chunks instead of assuming that removal leaves the remainder unchanged.

Offline closure-aware oracle analysis. The action grammar $\mathcal{A}$ contains target-span deletion, visible-sentence deletion, removal of all attack sentences, and document quarantine. For each action, ReIndexRepair performs four deterministic steps: (1) derive D_a^{aff} and $\Gamma(a)$; (2) construct $\widetilde{E}_a$; (3) run the complete retrieval-to-answer path in F_1; and (4) record the answer, dense top-30 IDs, served top-5 IDs, edited token count, and artifact hashes. After blinded human adjudication, the offline oracle selects

$$\begin{aligned} a_{\text{H}}^* &= \arg\min_{a \in \mathcal{A}} (c(a), \tau(a)) \\ \text{s.t.} \quad & h_t^{\text{H}}(Y_{F_2}) = 0, \quad h_c^{\text{H}}(Y_{F_2}) = 1, \quad \ell^{\text{H}}(Y_{F_2}) = 1, \end{aligned} \tag{5}$$

where $c(a)$ is removed-token cost and $\tau(a)$ is the lexicographic action name recorded in the frozen manifest, h_c^{H} is human correct-answer adoption, and ℓ^{H} is human completeness. If no candidate is feasible, the oracle records abstention. We materialize all four actions to measure same-query feasibility and cross-engine agreement; we do not claim a deployable semantic selector or online cost saving.

Equivalence certificate. *Proposition 1.* If F_1 and F_2 contain the same labeled vector multiset, share the query vector and pipeline fingerprint, and rank exact inner products using the same chunk-ID tie-break, then their dense rankings are identical; deterministic downstream stages yield identical served rankings and answers. To see this, each chunk has the same vector, label, and score in both multisets. Sorting the same score–ID pairs gives the same dense list, after which each downstream function receives the same ordered input. Our certificate checks chunk metadata, vector values at absolute tolerance 10^{-6}, exact top-30 IDs, exact top-5 served IDs, and answer

text for every action–incident pair. The tolerance is only a numerical vector check; rank identity is certified by the exact ID comparison. The result is query- and pipeline-specific and does not certify ANN topology, physical erasure, unseen questions, or model weights. The paths share frozen chunking, prompt, and generation helpers, so this is a separately materialized cross-engine equivalence check, not an independent implementation replication.

Protocol and failure handling. Incidents, fingerprints, grammar, hypotheses, and hashes are frozen first. Eligibility uses only initial and F_0 labels; trace-derived and complete-lineage F_2 endpoints are read afterward. The repair lattice separately compares F_1 with F_2; any hash, lineage, exact-view, completeness, safety, or correctness failure rejects the candidate. With N chunks, dimension d, and affected source D_a^{aff}, screening costs $C_{\text{embed}}(D_a^{\text{aff}}) + O(Nd) + O(N \log N) + C_{\text{rerank}}(k) + C_{\text{gen}}$ per action, excluding physical serialization.

Threat model. An attacker may inject several external documents but cannot modify model weights, trusted records, or the rebuild process. The defender receives a suspected span and its source provenance. We evaluate the repair logic under oracle provenance, then perturb locator recall and benign false positives separately. Source localization itself is outside our claim, and causal language is restricted to the frozen deterministic interventions in our constructed paired design.

State-transition invariants and audit object. The gate requires old and new descendant IDs to be disjoint, the unchanged-base fingerprint bundle to match, and the two materializations to agree in source revision, chunks, vectors, dense/served IDs, prompt hash, and answer. This distinguishes a closed repair from local deletion followed by incomplete refresh.

Run metadata hashes the corpus, cached vectors, scripts, and model trees; each action stores edit cost, dense/served IDs, answer, and equality checks. The pre-annotation manifest binds all 846 blinded items, while a separate versioned manifest binds the NumPy/FAISS certificate and server verifier without mutating that freeze. Verifiers check fingerprints, row alignment, coverage, and summaries and fail closed on inconsistent IDs. The complete-lineage arm is therefore a paired experimental control: it shares the question, generator, randomness policy, and rebuild implementation with the trace-derived arm, while changing only incident-provenance coverage. Within these frozen constructed interventions, this makes the endpoint the answer difference induced by provenance closure, rather than a comparison of unrelated defenses.

4. EXPERIMENTAL PROTOCOL

Frozen data. We use the 2,681,468-document BEIR Natural Questions corpus [15, 16]. Before retrieval outcomes were observed, a hash-locked certification set retained 94 mechanically valid synthetic incidents, each with five separately injected poison documents. The main generator experiment uses a 100k-document snapshot containing all relevant documents, published hard negatives, and hash-selected background documents. This is a controlled, hard-negative-enriched stress benchmark, not a prevalence estimate.

Pipeline and mechanisms. BGE-base-en-v1.5 [17] retrieves exact top-30, BGE-reranker-base serves top-5, and Qwen2.5-7B-Instruct [18] generates greedily with 96 new tokens. *Substitution* deletes only same-owner poison documents visible in the initial top-5; a fresh FAISS exact index [19] can promote hidden copies. *Boundary migration* uses a rule frozen before outcome inspection: one visible attack sentence is deleted from a constructed source, then the BGE tokenizer re-chunks at 128 tokens with zero overlap; a backup target sentence can move into a newly retrieved chunk. Both controls quarantine every descendant of the incident source and materialize a fresh index. Corpus, script, model-tree, and output hashes are recorded; a separate verifier recomputes IDs, lexical labels, and summaries, and checks root-manifest consistency.

Labels and statistics. The frozen flow is reported before conditioning. Target adoption, correct-answer adoption, and truncation follow a fixed rubric. Two independent humans labeled all 846 condition-blinded canonical outputs; 0 independent field decisions were `Unclear`, and a third person adjudicated 67 items. Raw agreement and three-class Cohen's κ were target adoption 96.7%/$\kappa = 0.90$, correct adoption 96.0%/$\kappa = 0.92$, completeness 98.8%/$\kappa = 0.88$. Final labels contain 43 incomplete outputs, all retained in the reported flow. Agreement uses Cohen's κ [20]. The two mechanism tests were frozen as the primary family in the hash-locked analysis manifest before human labels. We use two-sided exact McNemar tests [21], Holm correction across exactly these two tests [22], Clopper–Pearson intervals [23], and a paired 10,000-sample incident bootstrap [24] for the main risk difference. Eligibility depends only on the initial and F_0 labels, never either paired endpoint. For paired binary outcomes, the risk difference is $(n_{10} - n_{01})/n$; resampling keeps each incident's two outcomes together. Incomplete outputs remain in the denominator, with predeclared directional best/worst assignments reported as sensitivity analysis. These assignments vary only incomplete paired endpoints, holding adjudicated initial/F_0 eligibility fixed; they are not worst-case bounds over uncertain eligibility labels.

Audit and reproducibility gates. Every corpus snapshot, script, model tree, annotation source, and result has a recorded SHA-256 digest. A separate verifier reconstructs the 846 condition-blinded outputs, checks item coverage and immutable stimulus text, verifies positional cache alignment against the frozen corpus/query order, checks root-manifest consistency, and recomputes summary counts. It refuses to render paper macros unless the full hash chain and third-person decisions validate. Thus tables and figures share one generated statistics interface rather than manually copied numbers.

5. RESULTS

Two repair-mirage mechanisms. Table 2 reports every conditioning step. Substitution yields 13/71 recurrences after partial deletion versus 2/71 with provenance closure; the paired difference is 15.5 points (bootstrap 95% CI 7.0–23.9); unconditional F_2 target adoption is 15/94=16.0%. The exact 95% intervals of the two rates are 10.1–29.3% and 0.34–9.8%, respectively. Thus, even the control does not certify risk below 5% when its upper interval endpoint exceeds that threshold. Boundary migration yields 7/18 versus 1/18 (intervals 17.3–64.3% and 0.14–27.3%). Its unconditional F_2 target-adoption rate is 14/50=28.0%. Directional assignment of incomplete endpoints gives risk-difference ranges of substitution 5.6–21.1 points and boundary 16.7–38.9 points; neither least-favorable Holm test is significant ($p_{\text{Holm}} = .9062/.9062$).

Both mechanisms rebuild from source into a fresh exact index, so neither can be explained by stale vectors. In substitution, the removed top-5 evidence is absent after repair and recurrence comes from a previously unserved same-owner copy. In boundary migration, Independently of answer labels, migration occurs in 23/50 frozen constructions and 13/18 human-eligible cases; it accompanies 6/7 partial-arm recurrences. The lineage-complete arm is processed by the same code path, isolating provenance closure rather than rebuild policy.

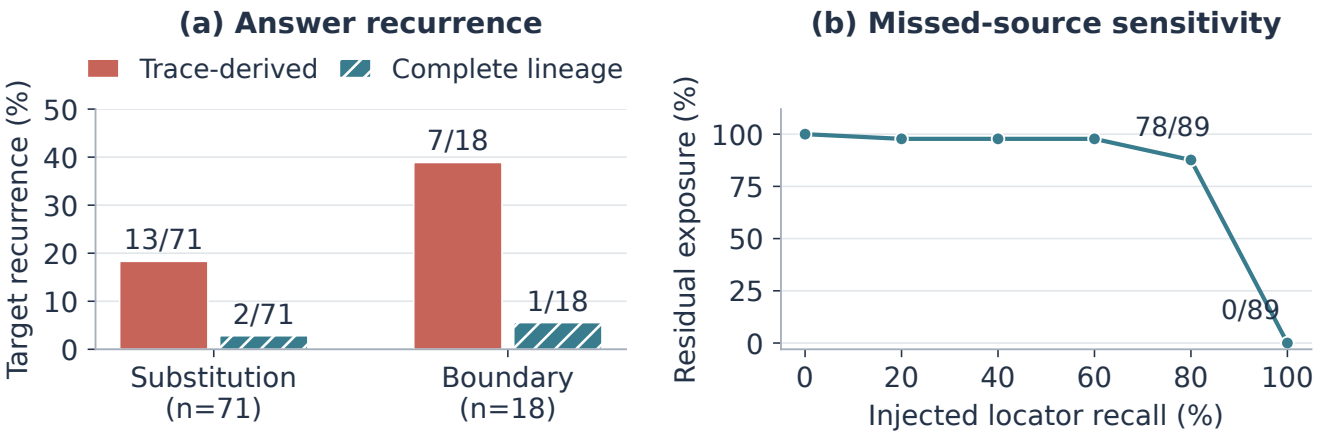


Fig. 2. Complementary answer- and retrieval-level evaluations. (a) Adjudicated F_2 target recurrence after an F_0-safe response. (b) Residual served-poison exposure among 89 initially exposed incidents under injected source recall with zero false positives.

Table 2. Recurrence after an F_0-safe response; flow is frozen $\rightarrow$ target adopted $\rightarrow$ F_0 safe.

Mechanism	Flow	Partial	Complete	p_{Holm}
Substitution	$94 \rightarrow 72 \rightarrow 71$	13/71	2/71	.0020
Boundary	$50 \rightarrow 20 \rightarrow 18$	7/18	1/18	.0312

Multiplicity and locator error. At served depth five, retrieval-level substitution for poison multiplicity $M = 1, 2, 3, 5$ is 0, 1.1, 11.4, and 52.8%; therefore the main phenomenon depends on attack redundancy and disappears at $M = 1$. In a separate deterministic sensitivity sweep, 89 incidents initially serve same-owner poison. Removing 0, 1, 2, 3, 4, or 5 of five sources leaves 89, 87, 87, 87, 78, or 0 incidents exposed (Fig. 2b). This isolates locator recall, not a deployed locator. A post-hoc 1M-document exact-retrieval run similarly promotes hidden poison in 49/88 initially exposed incidents (55.7%, exact 95% CI 44.7–66.3); it is a retrieval-only scale check, not additional answer-level evidence.

Utility. On 150 frozen non-incident queries, clean and lineage-complete corpora both retrieve a relevant document for 148/150 at top-30 and 121/150 at top-5. The poisoned corpus is 120/150 at top-5; repair produces one paired gain and no loss (change $+0.67$ points, bootstrap 95% CI 0–2.0). Deterministic false-positive removal of 100/100k benign documents changes no measured utility metric; 1000/100k causes one hit@5 loss (0.67 points). The counts preclude interpreting small decimals as broad gains.

Repair-lattice oracle analysis. On 30 boundary incidents, we exhaustively evaluate target-span, visible-sentence, all-attack-sentence, and document-quarantine actions on the triggering question. This is a feasibility and cost oracle, not held-out generalization. A safe and correct action exists for 19/30 (63.3% coverage); 11/30 abstain. For feasible cases, the least-token choices are 3 all-attack, 6 sentence, 10 span, with median 4.0 and mean 17.5 edited tokens. Separately materialized exact F_1 and fresh-FAISS F_2 paths agree for all 120 action–incident pairs in source chunks, dense and served IDs, prompt hashes, and answers; the maximum vector discrepancy is zero. All 270 initial/F_0/F_2 answers bound to human annotation remain byte-identical. Under the final human-adjudicated labels, exact-view oracle recall is 19/19 with zero token regret. All four F_2 actions were actually run for this offline oracle comparison; no deployable semantic selector or realized online rebuild saving is claimed. Correctness is part of feasibility: all 19 selected outputs retain it, versus 10/30 for document quarantine. These same-query figures must not be read as unseen-query ASR.

Evidence boundaries. Retrieved poison need not be adopted by the generator; unchanged hit@5 does not establish unchanged answer quality. Likewise, 120/120 exact-view matches certify tested questions and pipeline states, not removal from model parameters. The 11 abstentions are not repair successes. Hence recurrence, retrieval exposure, feasibility, and utility must be reported separately rather than inferred from one safe masked answer.

6. LIMITATIONS AND CONCLUSION

The study assumes oracle provenance, one QA corpus and generator, exact dense retrieval, and synthetic targets; the locator curve injects rather than measures errors. Finite certification is not universal safety, human judgments remain rubric-dependent, and the 30-case grammar is an oracle. Within these bounds, adjudicated estimates support lineage-complete matched controls for two constructed mechanisms; least-favorable incomplete-endpoint assignments retain positive differences but not significance. Exact counterfactual views provide a mechanically validated screening primitive for future deployable selectors when their equivalence is checked rather than assumed.

Experiments use public QA and synthetic targets, without real-person targeting.

7. ACKNOWLEDGMENTS

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8. REFERENCES

[1] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, 2020, vol. 33, pp. 9459–9474.

[2] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen tau Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781.

[3] Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia, "PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models," in *34th USENIX Security Symposium*, 2025, pp. 3827–3844.

[4] Tomohiro Yamashita, Daichi Amagata, and Yusuke Matsui, "How should we evaluate data deletion in graph-based ANN indexes?," in *NeurIPS Workshop on Machine Learning for Systems*, 2025.

[5] Baolei Zhang, Haoran Xin, Minghong Fang, Zhuqing Liu, Biao Yi, Tong Li, and Zheli Liu, "Traceback of poisoning attacks to retrieval-augmented generation," in *Proceedings of the ACM on Web Conference 2025*, 2025, pp. 2085–2097.

[6] Baolei Zhang, Haoran Xin, Yuxi Chen, Zhuqing Liu, Biao Yi, Tong Li, Lihai Nie, Zheli Liu, and Minghong Fang, "Who taught the lie? responsibility attribution for poisoned knowledge in retrieval-augmented generation," in *IEEE Symposium on Security and Privacy (SP)*, 2026, pp. 4203–4222.

[7] Huining Cui and Wei Liu, "Needle-in-rag: Prompt-conditioned character-level traceback of poisoned spans in retrieved evidence," *arXiv preprint arXiv:2605.01782*, 2026.

[8] Razieh Moradi, Havva Alizadeh Noughabi, Fattane Zarrinkalam, and Ali Dehghantanha, "Defending rag against knowledge poisoning using cross-encoder activation signals," in *Proceedings of the 39th Canadian Conference on Artificial Intelligence*, 2026, vol. 318 of *Proceedings of Machine Learning Research*, pp. 366–376.

[9] Pankayaraj Pathmanathan, Michael-Andrei Panaitescu-Liess, Cho-Yu Jason Chiang, and Furong Huang, "RAGPart & RAGMask: Retrieval-stage defenses against corpus poisoning in retrieval-augmented generation," *arXiv preprint arXiv:2512.24268*, 2025, AAAI 2026 Workshop on New Frontiers in Information Retrieval.

[10] Havva Alizadeh Noughabi, Fattane Zarrinkalam, and Ali Dehghantanha, "Defense against knowledge poisoning attack on GraphRAG," in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 2026, pp. 555–563.

[11] Zhiyuan Chang, Mingyang Li, Xiaojun Jia, Junjie Wang, Yuekai Huang, Ziyou Jiang, Yang Liu, and Qing Wang, "One shot dominance: Knowledge poisoning attack on retrieval-augmented generation systems," in *Findings of the Association for Computational Linguistics: EMNLP*, 2025, pp. 18811–18825.

[12] Xi Nie, Hongwei Li, Shenghao Wu, Mingxuan Li, Jiachen Li, and Wenbo Jiang, "When poison fails after retrieval: Revisiting corpus poisoning under chunking and reranking pipelines," *arXiv preprint arXiv:2606.11265*, 2026.

[13] Shang Wang, Tianqing Zhu, Dayong Ye, and Wanlei Zhou, "When machine unlearning meets retrieval-augmented generation (RAG): Keep secret or forget knowledge?," *IEEE Transactions on Dependable and Secure Computing*, vol. 23, no. 2, pp. 2030–2045, 2026.

[14] Yujing Wang, Jinwen Chen, Hainan Zhang, Liang Pang, Yongxin Tong, Binghui Guo, Hongwei Zheng, and Zhiming Zheng, "Learning to erase private knowledge from multi-documents for retrieval-augmented large language models," *arXiv preprint arXiv:2504.09910*, 2025, Revised June 2026.

[15] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych, "BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021, vol. 1.

[16] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov, "Natural questions: A benchmark for question answering research," *Transactions of the Association for Computational Linguistics*, vol. 7, pp. 453–466, 2019.

[17] Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie, "C-Pack: Packed resources for general chinese embeddings," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 641–649.

[18] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, et al., "Qwen2.5 technical report," *arXiv preprint arXiv:2412.15115*, 2024, Revised January 2025.

[19] Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou, "The Faiss library," *arXiv preprint arXiv:2401.08281*, 2024, Revised October 2025.

[20] Jacob Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.

[21] Quinn McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[22] Sture Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[23] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.

[24] Bradley Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.